# Project Report

## Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau


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# 1. Introduction

## 1.1 Project Overview

Cosmetic Insights is an interactive Tableau dashboard built to help cosmetics brand managers, analysts, and business leadership understand market trends and consumer behavior. It analyzes a cosmetics dataset covering brand, product category, price, rank, and skin-type suitability (Dry, Oily, Combination, Normal, Sensitive) to surface brand performance, pricing patterns, and skin-type coverage gaps.

## 1.2 Purpose

Brand and category managers currently rely on manually compiled spreadsheets scattered across brand, pricing, and ranking data, with no simple way to connect that information to skin-type suitability. This makes pricing, marketing, and product-mix decisions slow and inconsistent. Cosmetic Insights solves this by giving teams one connected, filterable view they can use to make faster, evidence-backed decisions.

# 2. Ideation Phase

## 2.1 Problem Statement

| PS | I am | I'm trying to | But | Because | Which makes me feel |
|---|---|---|---|---|---|
| PS-1 | a Brand & Category Manager | identify top products/price segments by skin type | data sits in disconnected sheets | no single dashboard connects it | overwhelmed, unsure decisions are evidence-based |
| PS-2 | a Data/Market Analyst | spot emerging consumer preference trends | I must clean/cross-reference data manually | data isn't pre-processed or visualized | slowed down, behind on delivering insights |
| PS-3 | a shopper | find products suited to my skin type at a fair price | rank/skin-suitability info is hard to compare | no simple visual way to filter and compare | uncertain, likely to buy the wrong product |

## 2.2 Empathy Map Canvas

Persona: Priya, a Brand & Category Manager at a mid-size cosmetics retail company, responsible for pricing, product mix, and marketing spend across many brands and skin-type segments.

- Says/Does: manually compiles reports from multiple spreadsheets; cross-checks competitor pricing by hand
- Thinks/Feels: worried about mispricing a product; frustrated data lives in separate places
- Sees/Hears: a crowded, fast-moving market; leadership asking for justification of spend
- Pain: no single reliable view connecting brand, price, rank, and skin type
- Gain: one interactive dashboard to explore all of it together, faster evidence-backed decisions

## 2.3 Brainstorming

The team selected PS-1 as the primary problem to solve, with PS-2 as a supporting data-readiness workstream. Ideas were grouped into: Data Preparation & Preprocessing, Brand & Pricing Analysis, Skin-Type & Category Insights, Dashboard Design & Interactivity, and Predictive & Advanced Insights. Prioritization (impact vs. feasibility) placed dataset cleaning, core brand/price/rank views, and skin-type filters as “Do Now” items, with predictive trend indicators and competitor benchmarking as “Do Next.”

# 3. Requirement Analysis

## 3.1 Customer Journey Map

Priya's journey runs from Awareness (frustrated with manual reporting) through Consideration, Onboarding, Exploration, Decision-Making, and Advocacy — moving from frustration and uncertainty to confidence and trust as she adopts the dashboard for every review cycle.

### 3.2 Solution Requirement

Functional requirements cover data upload & preprocessing, brand & pricing analysis, skin-type segmentation, interactive dashboard & filters, user access, and trend/insight reporting. Non-functional requirements cover usability, security, reliability, performance, availability, and scalability.

### 3.3 Data Flow Diagram

Level 0: Data Analyst → Cosmetic Insights Dashboard System → Brand Manager/Leadership. Level 1 breaks this into (1) Data Cleaning & Preprocessing feeding a Cosmetics Dataset store, (2) a Brand, Price & Skin-Type Analysis Engine reading that store into (3) an Interactive Tableau Dashboard that serves filtered views and reports back to users.

### 3.4 Technology Stack

| Component | Technology |
| --- | --- |
| User Interface | Tableau Desktop, Tableau Public/Server/Cloud |
| Data Source | CSV / Excel cosmetics dataset |
| Data Preprocessing | Tableau Prep, Python (Pandas), Excel |
| Application Logic | Tableau Calculated Fields, LOD Expressions, Filters, Actions, Parameters |
| Hosting / Publishing | Tableau Public, Server, or Cloud |

## 4. Project Design

### 4.1 Problem Solution Fit

Customer segment: Brand/Category Managers, Analysts, and Leadership at cosmetics companies. Their jobs-to-be-done — identifying top performers, spotting skin-type gaps, and justifying pricing/marketing decisions — are currently done through manual spreadsheet work triggered by upcoming reviews or competitor moves. The proposed solution, Cosmetic Insights, is delivered online via Tableau Cloud/Server links and offline via exported reports.

### 4.2 Proposed Solution

An interactive Tableau dashboard that ingests and cleans the cosmetics dataset, then visualizes brand performance, pricing trends, and skin-type coverage through filterable, drill-down views. Its novelty is connecting brand, price, rank, and skin-type suitability in one view rather than the separate spreadsheets most teams use today. It can be used as an internal BI tool or offered as a Tableau-based template/subscription to smaller cosmetics brands.

### 4.3 Solution Architecture

Cosmetics Dataset (CSV/Excel) → Data Preparation (Tableau Prep/Python) → Data Model (brand, price, rank, skin type) → Analytics & Visualization (Tableau Desktop) → Publish (Tableau Public/Server/Cloud) → Consumers (Analysts, Brand Managers, Leadership).

## 5. Project Planning & Scheduling

### 5.1 Project Planning

The product backlog covers 10 user stories (USN-1 to USN-10) across two sprints. Sprint-1 focused on data upload/preprocessing, initial brand & pricing views, and user access (16 story points). Sprint-2 covered skin-type segmentation, interactive drill-down, and trend/export reporting (16 story points). Total story points = 32 across 2 sprints, giving the team a velocity of 16 story points per sprint.

## 6. Functional and Performance Testing

### 6.1 Performance Testing

Tableau performance testing confirmed the cosmetics dataset rendered correctly with all fields typed properly; preprocessing standardized brand names, skin-type labels, and price bands; global and quick filters worked across brand, category, price, and skin type; and calculated fields (Price Band, Rank Tier, Price-to-Rank Ratio) computed as expected. The dashboard shipped with 6 core visualizations and a 3-part story. User Acceptance Testing ran 6 test cases (dataset upload, skin-type filter, drill-down, price filter, export, role-based access) — all passed — with 2 minor UI bugs found and resolved.

## 7. Results

### 7.1 Output Screenshots

[Insert screenshots of the Brand & Pricing Analysis view, Skin-Type Coverage heatmap, interactive filters, and the summary Story here once the workbook is finalized.]

## 8. Advantages & Disadvantages

***Advantages:***

- Brings brand, price, rank, and skin-type data into one connected, filterable view
- Cuts down hours of manual spreadsheet reporting before every review
- Makes skin-type coverage gaps and pricing outliers easy to spot
- Built on Tableau, so it's approachable for teams with no dedicated BI engineers

***Disadvantages:***

- Depends on the quality and freshness of the uploaded dataset
- Descriptive/exploratory analytics only — no predictive modeling in the current version
- Tableau Cloud/Server licensing is an ongoing cost for smaller teams

## 9. Conclusion

Cosmetic Insights gives cosmetics brand managers, analysts, and leadership a single, interactive view of brand, price, rank, and skin-type data — replacing slow, manual, spreadsheet-based reporting with fast, filterable, evidence-based insight. Testing across performance and user acceptance criteria confirmed the dashboard meets its core requirements and is ready for use in pricing and marketing review cycles.

## 10. Future Scope

- Predictive trend indicators for emerging categories and skin-type demand shifts
- Automated alerts when consumer preference or pricing shifts significantly
- Competitor benchmarking view once external pricing data is available
- Sentiment analysis of customer reviews to complement the ranking data
- Mobile-friendly version of the dashboard for on-the-go review

## 11. Appendix

***Source Code (if any):***

Tableau workbook (.twbx) and any Tableau Prep flows / Python preprocessing scripts used to clean the dataset.

***Dataset Link:***

[Add the link to the cosmetics dataset used for this project.]

***GitHub & Project Demo Link:***

[Add the GitHub repository link and the published Tableau Public/Cloud demo link here.]